

PRANVEER SINGH INSTITUTE OF TECHNOLOGY

B.Tech Computer Science — Semester Examination Simulation

SUBJECT: DESIGN ANALYSIS OF ALGORITHM

Paper Set:	Set 02	Duration:	3 Hours
Maximum Marks:	100 Marks	Date:	September 2026

SECTION A

Answer all questions in brief. (10 x 2 = 20 Marks)

- Q1(a).** Write down the basic properties or conditions required for Analyzing Algorithms. [2M]
- Q1(b).** Differentiate in brief between Heap Sort and Complexity of Algorithms. [2M]
- Q1(c).** List any two advantages and limitations of Complexity of Algorithms. [2M]
- Q1(d).** Give a real-world example illustrating Growth of Functions. [2M]
- Q1(e).** Define Introduction and state its key application in computer systems. [2M]
- Q1(f).** What is the primary significance of Algorithms? Explain in brief. [2M]
- Q1(g).** State the fundamental principles governing Comparison of Sorting Algorithms. [2M]
- Q1(h).** Discuss the key methodologies and implementations of Analyzing Algorithms. [2M]
- Q1(i).** Discuss the key methodologies and implementations of Heap Sort. [2M]
- Q1(j).** Discuss the key methodologies and implementations of Complexity of Algorithms. [2M]

SECTION B

Attempt any three of the following questions. (3 x 10 = 30 Marks)

- Q2.** Design a solution using Growth of Functions for an enterprise system and trace each step. [10M]
- Q3.** Explain the theoretical foundation of Introduction and prove its correctness. [10M]
- Q4.** Write a step-by-step algorithm or procedure for Algorithms with illustrative inputs. [10M]
- Q5.** Discuss the various challenges encountered in Comparison of Sorting Algorithms and how they are mitigated. [10M]
- Q6.** Explain Analyzing Algorithms in detail with suitable architectural diagrams and examples. [10M]

SECTION C

Attempt any five questions from this section. (5 x 10 = 50 Marks)

- Q5.** Design a solution using Heap Sort for an enterprise system and trace each step. [10M]
- Q6.** Explain the theoretical foundation of Complexity of Algorithms and prove its correctness. [10M]
- Q7.** Write a step-by-step algorithm or procedure for Growth of Functions with illustrative inputs. [10M]
- Q8.** Discuss the various challenges encountered in Introduction and how they are mitigated. [10M]
- Q9.** Explain Algorithms in detail with suitable architectural diagrams and examples. [10M]
- Q10.** Discuss the algorithmic approach for Comparison of Sorting Algorithms. Analyze its time and space complexity. [10M]

Q11. Compare and contrast Analyzing Algorithms with Heap Sort. Highlight performance trade-offs.

[10M]

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